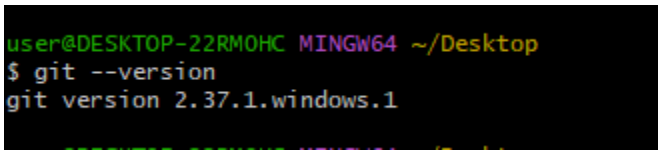
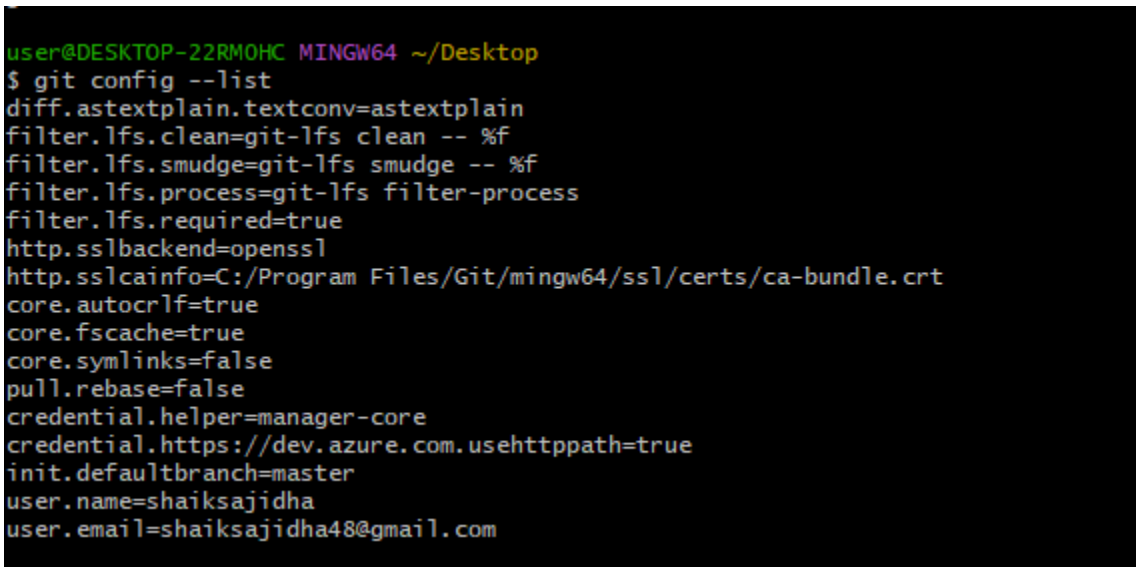
1. Installation of Git



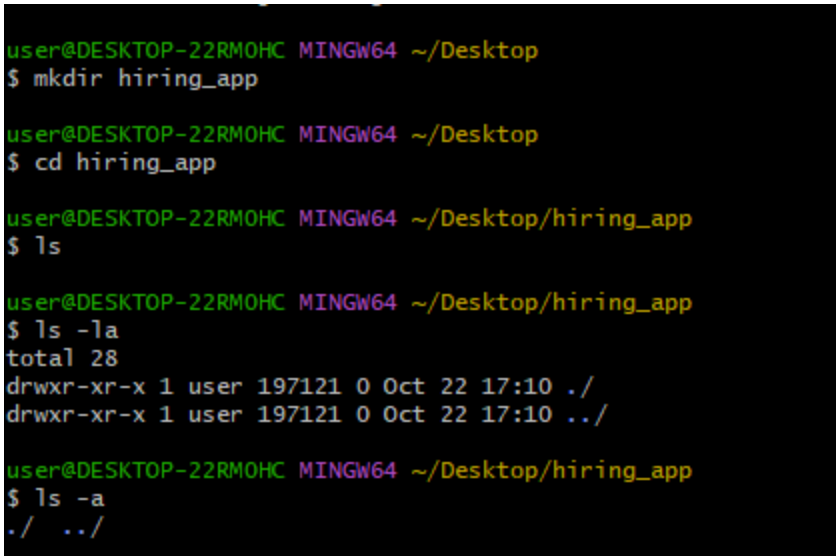
I used sudo apt update to update the package information. Then I used sudo apt install git -y to install Git. Finally, I used git --version to verify that Git was installed successfully.

2. Configure Git



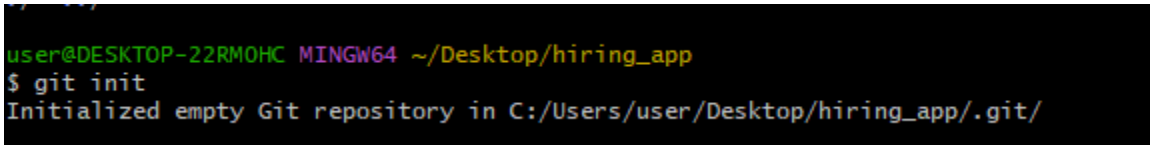
I used git config --global user.name to configure my Git username. I used git config --global user.email to configure my email address. These details are associated with the commits I create. I used git config --list to verify my configuration.

3. Created a project directory hiring_app



I used mkdir hiring-app to create my project directory. Then I used cd hiring-app to enter the directory. I used pwd to check my current location and ls to see the files. I used ls -la to also display hidden files.

4. Initialize Git



I used git init to initialize Git in my project directory. This created a hidden .git directory, which stores Git's repository information.

5. Create files

I used touch 1 2 3 4to create four files. Then I used git status to check the repository. Git showed these files as untracked because I had not yet added them to Git's staging area.

```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ touch 1 2 3 4

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ ls
1  2  3  4

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        1
        2
        3
        4

nothing added to commit but untracked files present (use "git add" to track)

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$
```

6. Adding files to tracking

```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git add .

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git status
On branch master

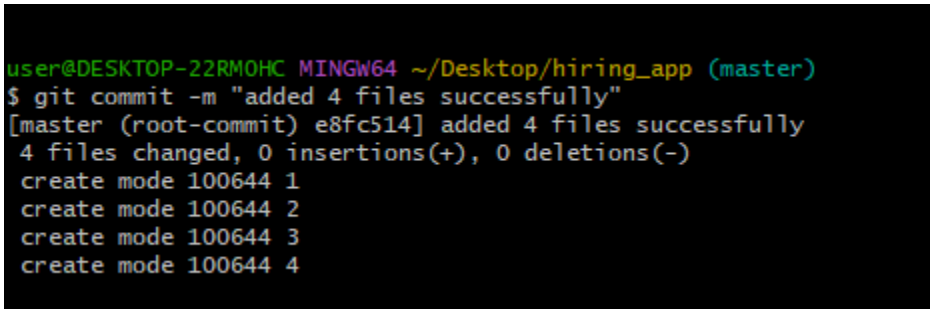
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   1
        new file:   2
        new file:   3
        new file:   4


user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$
```

I used git add to move files from the working directory to the tracking area. I can add one file, multiple files, or all files using git add .. Then I used git status to verify which files were staged.

7. Commit the changes



I used git commit to permanently save the staged changes in the Git repository. I provided "added 4 files successfully" as the commit message so I could understand what changes were made.

8. Check commit history

```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git log
commit e8fc5147422ec4ec0e3b4641bbf289f087754e12 (HEAD -> master)
Author: shaiksajidha <shaiksajidha48@gmail.com>
Date:   Thu Oct 22 17:41:24 2026 +0530

    added 4 files successfully

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git log -4
commit e8fc5147422ec4ec0e3b4641bbf289f087754e12 (HEAD -> master)
Author: shaiksajidha <shaiksajidha48@gmail.com>
Date:   Thu Oct 22 17:41:24 2026 +0530

    added 4 files successfully

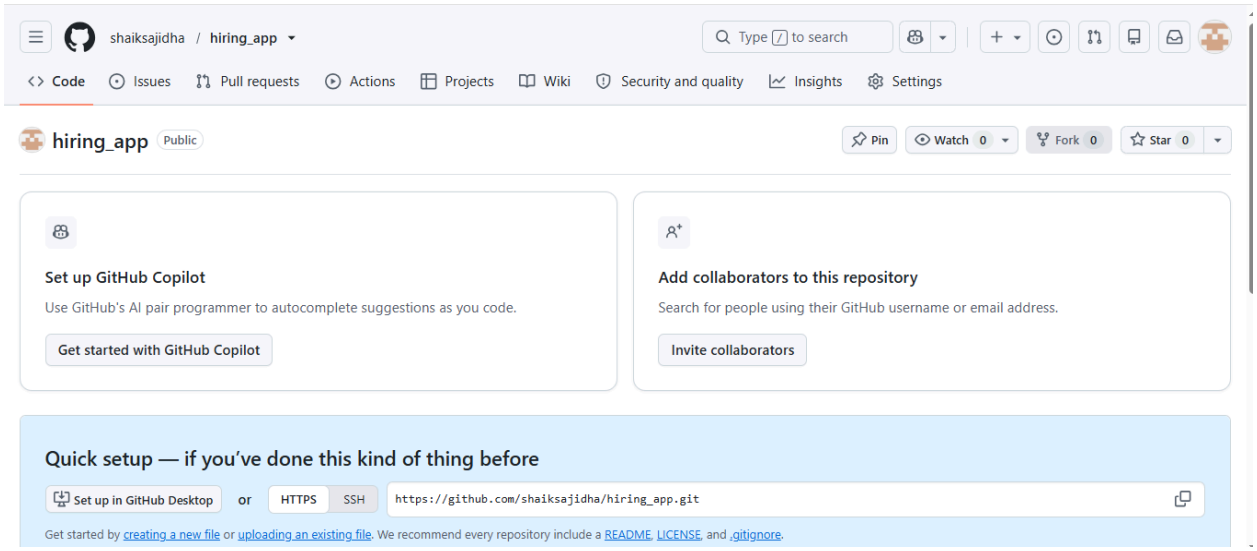
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git log --online
fatal: unrecognized argument: --online

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git log --oneline
e8fc514 (HEAD -> master) added 4 files successfully

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$
```

I used git log to view the complete commit history. I used git log --oneline to see the commits in a short format. I used git log -4 to display the latest four commits.

9. Create GitHub repository and connect it



```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git remote add origin git@github.com:shaiksajidha/hiring_app.git

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git remote -v
origin  git@github.com:shaiksajidha/hiring_app.git (fetch)
origin  git@github.com:shaiksajidha/hiring_app.git (push)

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$
```

I created a repository named hiring-app on GitHub. Then I used git remote add origin to connect my local Git repository with the GitHub repository. I used git remote -v to verify the remote URL.

10. Rename branch to main

```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git branch
* master

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (master)
$ git branch -m owner

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$ git branch
* owner

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$
```

git branch
git branch --show-current
git branch -M main

I used git branch to check my current branch. I used git branch --show-current to display the current branch name. Then I used git branch -M owner to rename my branch to main.

11. Push code to GitHub

```
user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ git push -u origin owner
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ ls -la ~/.ssh
total 28
drwxr-xr-x 1 user 197121    0 Oct 14 18:22 ./
drwxr-xr-x 1 user 197121    0 Oct 22 17:58 ../
-rw-r--r-- 1 user 197121 1969 Oct 22 18:03 known_hosts
-rw-r--r-- 1 user 197121 1566 Oct 14 18:22 known_hosts.old

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/user/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/user/.ssh/id_rsa
Your public key has been saved in /c/Users/user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:D/sXBPhIJOQVXS6KAnTqBH3Q6qGBdwjqRsgB9St2cz8 user@DESKTOP-22RM0HC
The key's randomart image is:
+---[RSA 3072]----+
|+o+o..o.++ ..    |
|.+.+o. oo o.     |
|=.=+. .. o...    |
|===... ......    |
|o=+o= o S  .     |
|.+.o + . +  .    |
|.       E .  .   |
|         o  .    |
|          ..     |
+----[SHA256]-----+

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ ls -la ~/.ssh
total 33
drwxr-xr-x 1 user 197121    0 Oct 22 18:06 ./
drwxr-xr-x 1 user 197121    0 Oct 22 17:58 ../
```

```
user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ ls -la ~/.ssh
total 33
drwxr-xr-x 1 user 197121    0 Oct 22 18:06 ./
drwxr-xr-x 1 user 197121    0 Oct 22 17:58 ../
-rw-r--r-- 1 user 197121 2610 Oct 22 18:06 id_rsa
-rw-r--r-- 1 user 197121  574 Oct 22 18:06 id_rsa.pub
-rw-r--r-- 1 user 197121 1969 Oct 22 18:03 known_hosts
-rw-r--r-- 1 user 197121 1566 Oct 14 18:22 known_hosts.old

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ ^C

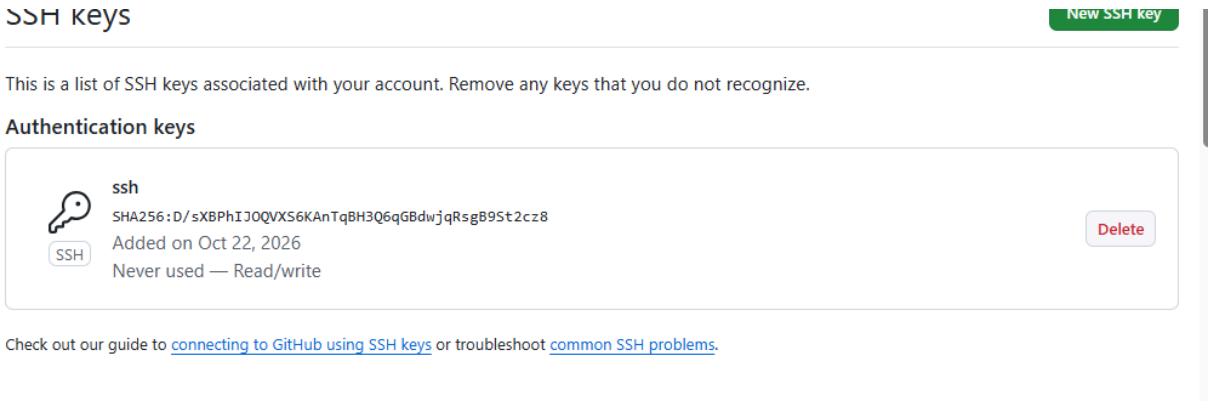
user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ cat id_rsa.pub
cat: id_rsa.pub: No such file or directory

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCnAwilkjZVH2ybOrVXtMCgc6M5LgUwUKelil9+J84m
Sv/gb6BOIsVoiofjS7t57ojmLCzAJ7om2lk7rQFIBShxq0bTZ/5TOIExE2mGUsg+sa9OaywdDEwhlTfh
s14+ESAZrRdfT1q2xedDDR7GHN4g95ikMx62T33BMsEwrR8OARwKHBv+u2UbLShfMSFOJuKPJiQ/Jkia
VRlqallHpuWVO5F4e8JoDCcvT7aHcti7OaiQVtnSacRv1xst9P3PLfG216VAOWrZv+O5dYYtBdAtOUZB
XlijkSqvXWHMGHTWi7A/fkp9N7J6Kqv6Pit/L6GCoa1H/6BFgxqQndKU/1dY99w+/ib83VCAdYOjM18i
qDL1/TkFUYXzjUmJVI3S3F2/YvWIv1w2tqP/JhFZOXwX6/zdAtNO4+6oKh2p4+xwOXbaF/aNNUsTNWV0
NgN5oyGxTa9Ylt+v8jyqvOvF+3AVIkCO+uxXBuASHwq0KKMyEnOoiFlddDquPQIr/EWAYuE= user@DE
SKTOP-22RM0HC

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ git push -u origin owner
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 225 bytes | 112.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:shaiksajidha/hiring_app.git
 * [new branch]      owner -> owner
branch 'owner' set up to track 'origin/owner'.

user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$ git push
Everything up-to-date

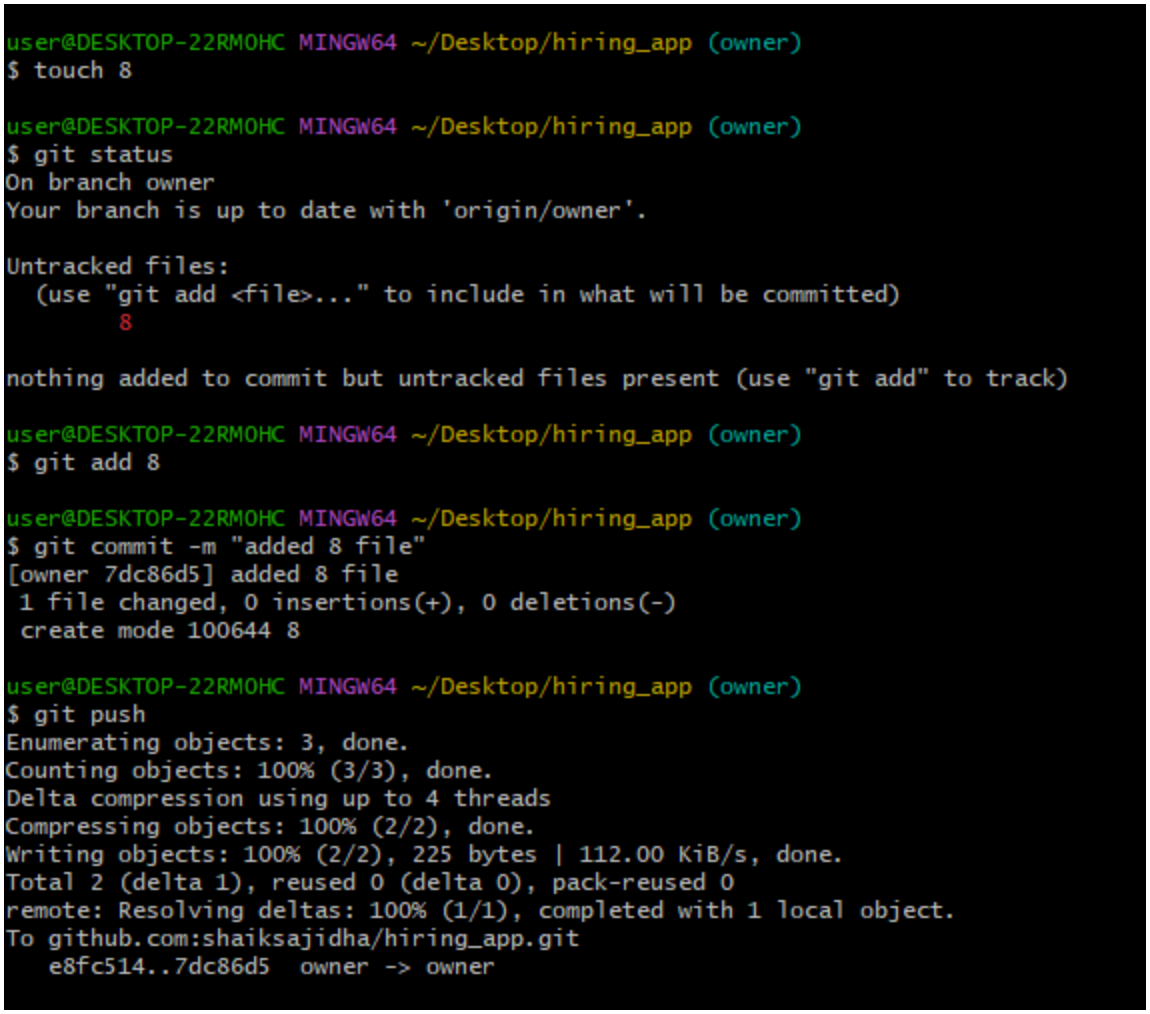
user@DESKTOP-22RM0HC MINGW64 ~/Desktop/hiring_app (owner)
$
```



git push -u origin main
I used git push -u origin main to upload my local commits to the GitHub repository. The origin represents my remote GitHub repository, and main represents the branch I want to push. I used git push to upload my subsequent local commits to GitHub.

12. Make another change.

touch D
git status
git add D
git commit -m "Added D file"
git push
I created a new file called D. I used git status to check the change. Then I used git add D to stage the file, git commit to save the change, and git push to upload the new commit to GitHub.



13. Pull changes from GitHub

git pull

I used git pull to download the latest changes from the remote GitHub repository and integrate those changes into my current local branch.

```
   e8fc514..7dc86d5  owner -> owner

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 908 bytes | 50.00 KiB/s, done.
From github.com:shaiksajidha/hiring_app
   7dc86d5..b8731bd  owner      -> origin/owner
Updating 7dc86d5..b8731bd
Fast-forward
 saji | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 saji

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$ git status
On branch owner
Your branch is up to date with 'origin/owner'.

nothing to commit, working tree clean
```

14. Fetch changes

git fetch
git fetch origin

I used git fetch to download information about changes from the remote repository without automatically merging those changes into my current branch.

Easy difference:
- git fetch → downloads changes but does not merge.
- git pull → downloads changes and integrates them.

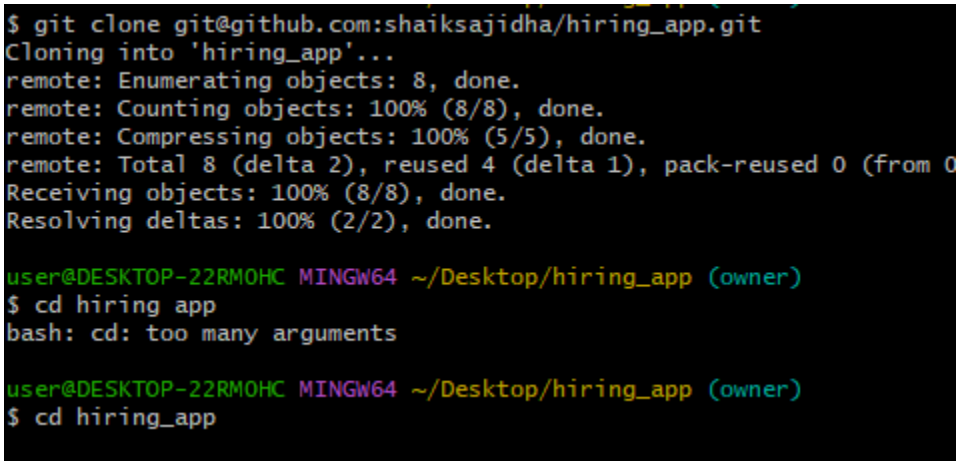
```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$ git fetch

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$ git fetch origin

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app (owner)
$
```

15. Clone a repository

git clone git@github.com:YOUR_USERNAME/hiring-app.git
cd hiring-app
I used git clone when I wanted to get an existing GitHub repository onto my local machine. It downloads the repository, including its files and Git history.



16. Unstage a file

git reset F

I used git reset F when I wanted to remove file F from the staging area without deleting the file from my working directory.

```
user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ touch f

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ ls
1  2  3  4  8  f  saji

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ git status
On branch owner
Your branch is up to date with 'origin/owner'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        f

nothing added to commit but untracked files present (use "git add" to track)

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ git add f

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ git status
On branch owner
Your branch is up to date with 'origin/owner'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   f


user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ git reset f

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$ git status
On branch owner
Your branch is up to date with 'origin/owner'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        f

nothing added to commit but untracked files present (use "git add" to track)

user@DESKTOP-22RMOHC MINGW64 ~/Desktop/hiring_app/hiring_app (owner)
$
```